AI-Powered Multi-Label Food Ingredient Recognition with Uncertainty Quantification

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Background

Problem Statement:

- Manual food logging is time-consuming and inaccurate
- Nutritional tracking requires precise ingredient identification
- Existing methods lack confidence quantification

Challenge:

- Multi-label classification (avg 10 ingredients per dish)
- 249 ingredient classes with severe class imbalance
- Need reliable predictions for health applications

Motivations & Objectives

Motivation:

- Automatic dietary tracking for health management
- AI-assisted nutrition analysis
- Scalable to mobile applications

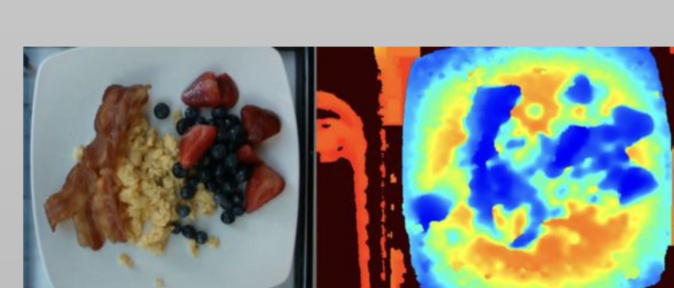
Objectives:

1. Build accurate multi-label ingredient detector
2. Quantify prediction uncertainty for reliability
3. Handle class imbalance effectively
4. Provide nutritional insights from predictions
5. Create deployable demo system

Dataset & Parameters

Dataset: Nutrition5K

- 3,490 RGB + Depth images
- 2,792 train / 349 val / 349 test
- 249 ingredient classes
- Multi-label annotations



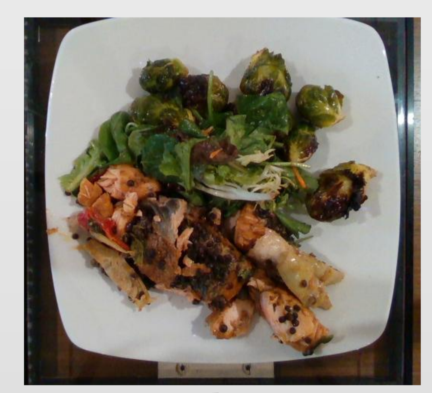
Training Strategy:

- Model: EfficientNet-B3 (16.3M params)
- Optimizer: AdamW (lr=1e-3, weight_decay=1e-5)
- Batch Size: 32 | Epochs: 20
- Augmentation: Flip, Rotation, ColorJitter
- Regularization: Dropout (0.3, 0.15) + L2
- Hardware: V100 GPU | Training time: 45 min

Methodology

Model Architecture:

RGB Image (512*512)



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EfficientNet-B3 backbone (ImageNet pretrained)

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1536 features

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Classifier Dropout + MLP

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BCEwithLogitsLoss + MFB weights

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249-dim predictions

Key Innovations:

A. MFB Class Weighting

Assign higher weights to rare ingredients (up to 50×) using Weight_i = Median_Freq / Class_Freq_i to handle class imbalance.

B. Threshold Optimization

Systematically test thresholds 0.10-0.85; optimal at **0.20** improves F1 by 1.6% over default 0.5.

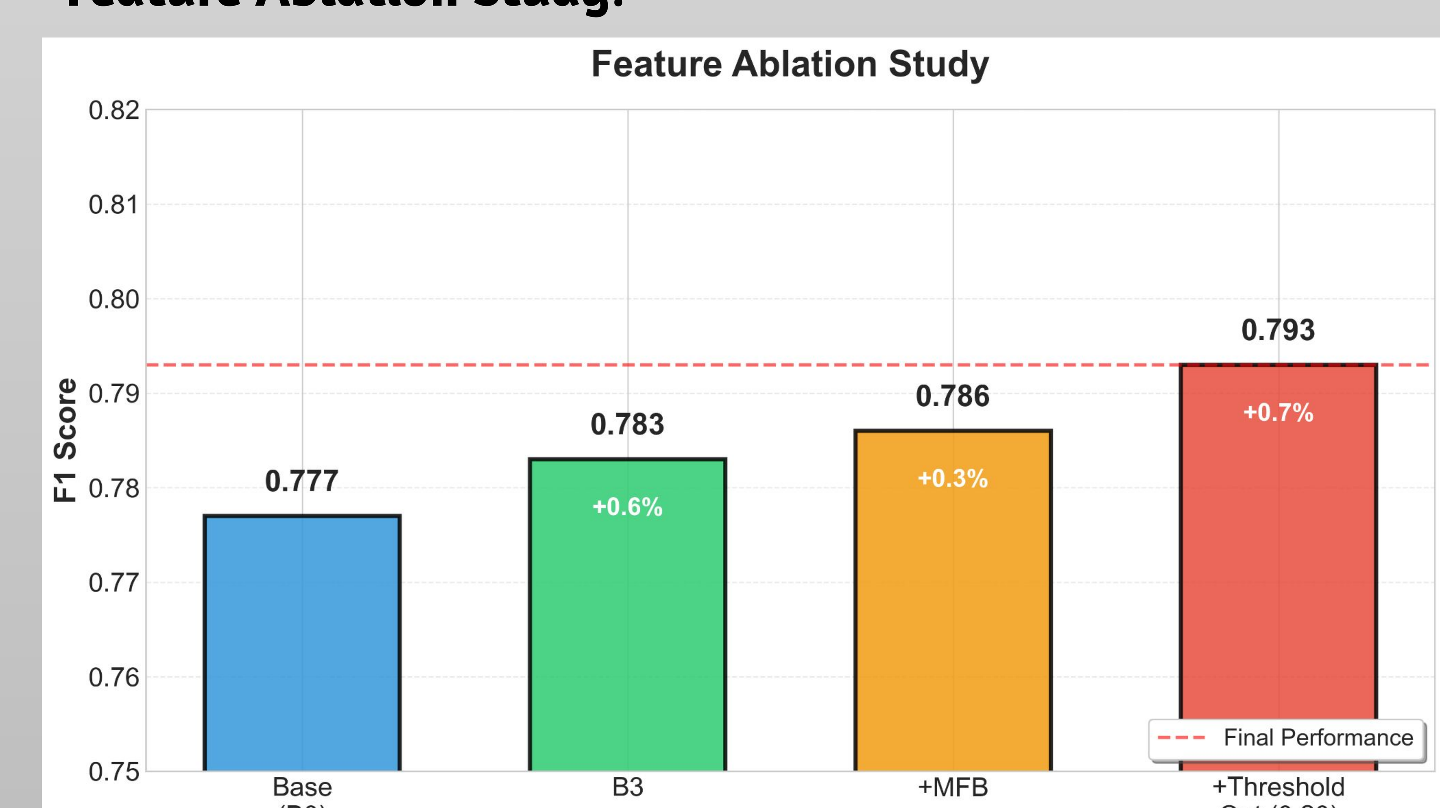
C. MC Dropout Uncertainty

Run 20 forward passes with dropout enabled; calculate mean ± std to quantify prediction confidence.

D. Co-occurrence Learning (Tested)

Learn 6,999 ingredient pairs from training data and test boosting strategy; found that it decreased F1 by 7.6% due to false positives, thus not adopted in final system.

Feature Ablation Study:

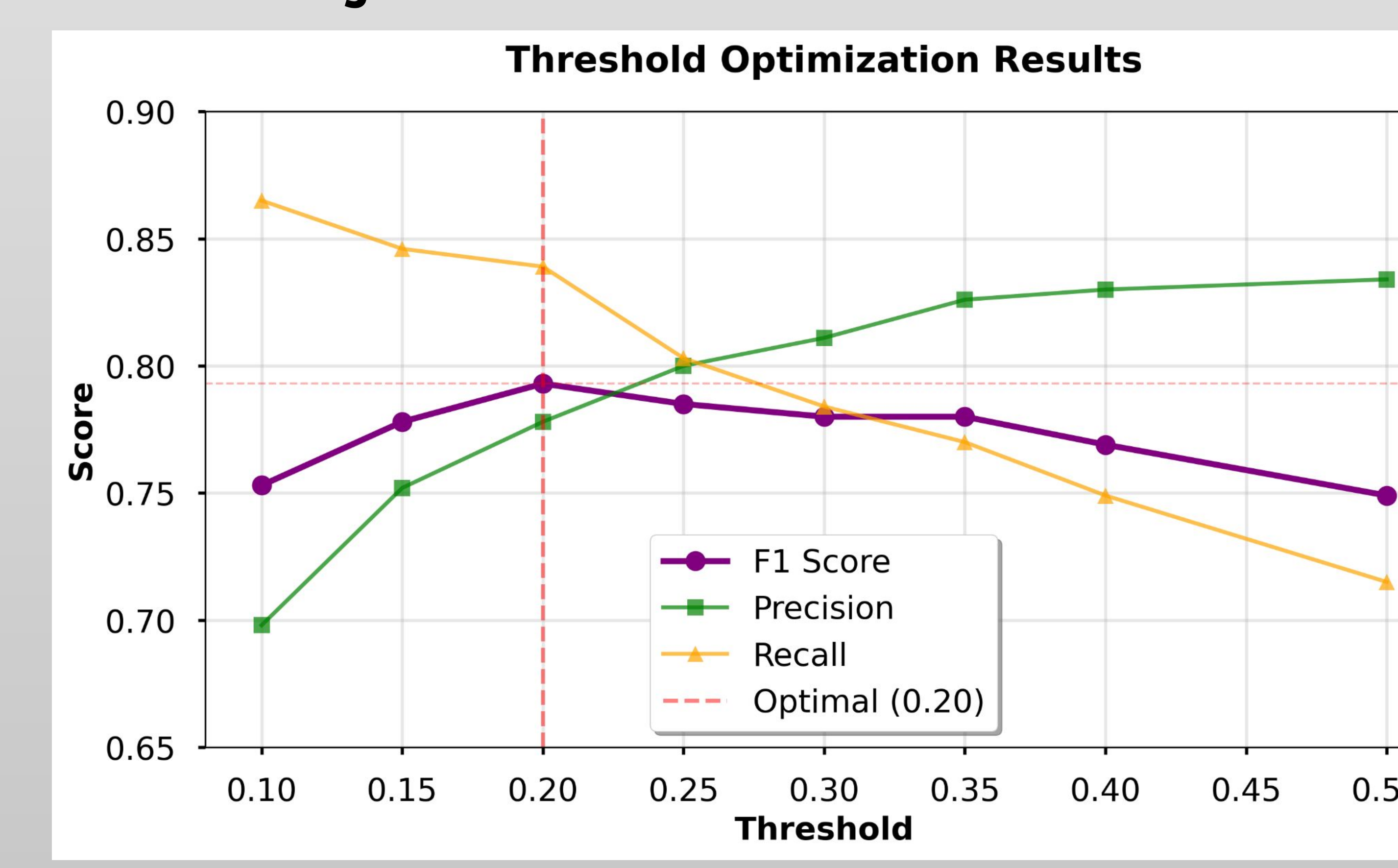


Results

Performance Metrics:

Metric	Value
F1 Score	0.793
Precision	0.778
Recall	0.839
Exact Match	0.364
Hamming Accuracy	0.988

Threshold Analysis:



Conclusion

Conclusion:

- Achieved 0.793 F1 score on 249-class multi-label task
- MFB weighting effectively handles class imbalance
- Uncertainty quantification provides reliability scores
- End-to-end system from image to nutrition advice

Key Contributions:

1. Systematic threshold optimization improving F1 by 1.6%
2. MC Dropout for uncertainty in food recognition
3. Deployable Streamlit application

Scan to see the app!

Reference

[1] Siddhant. (2025, April 16). *Nutrition5k: A comprehensive nutrition dataset*. Kaggle. https://www.kaggle.com/datasets/siddhantrout/nutrition5k-dataset/data
[2] Thames, Q., Karpur, A., Norris, W., Xia, F., Panait, L., Weyand, T., & Sim, J. (2021b, June 22). *Nutrition5k: Towards automatic nutritional understanding of generic food*. arXiv.org. https://arxiv.org/abs/2103.03375